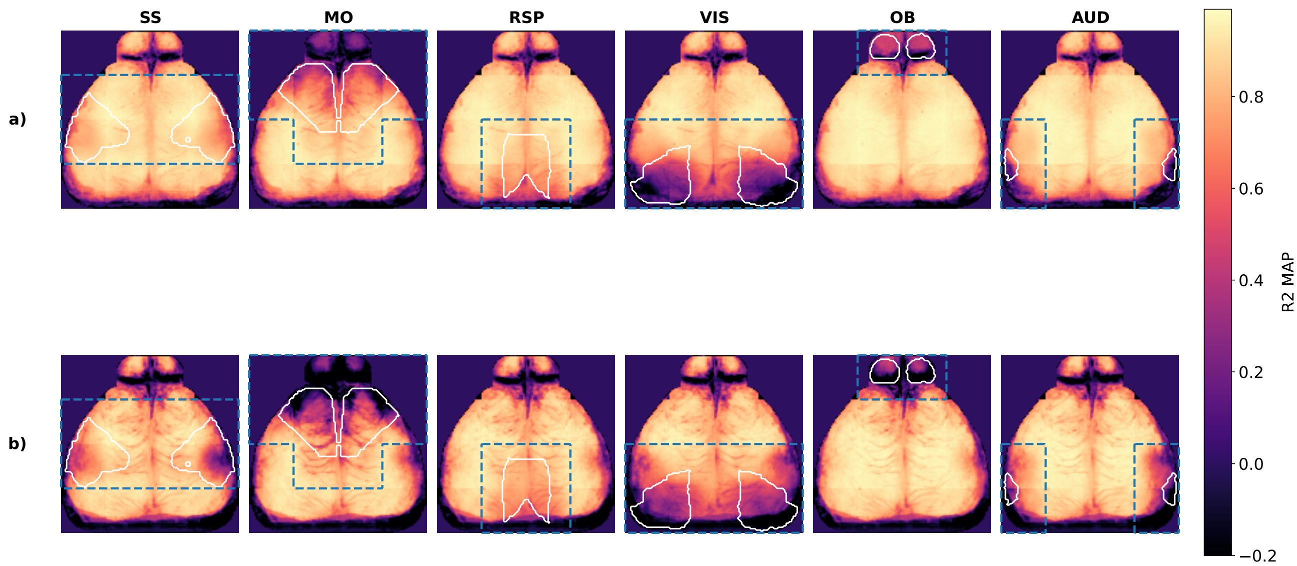

Figure 14. **Per-pixel R^2 maps for left-out region prediction.** Per-pixel R^2 for **(a)** held-in subjects (top row; pretraining set) and **(b)** held-out subjects (bottom row; finetune and zero-shot sets), averaged across all sessions. Each column corresponds to masking the region bounded by the blue dashed lines and predicting its activity from the rest of the brain as context. The white boundaries in each panel illustrate the target ROI. **Evaluation for each column focuses on the corresponding masked ROI**, where the ROI and its neighboring regions are removed to eliminate local spatial correlations, ensuring that performance reflects cross-region prediction rather than trivial neighborhood interpolation. A higher R^2 within the masked region indicates successful cross-region prediction. The results show that WiCAT maintains strong predictive performance across both seen and unseen subjects, with performance largely preserved when transitioning to held-out subjects, indicating that global spatial patterns are learned and transferred effectively to the zero-shot set.

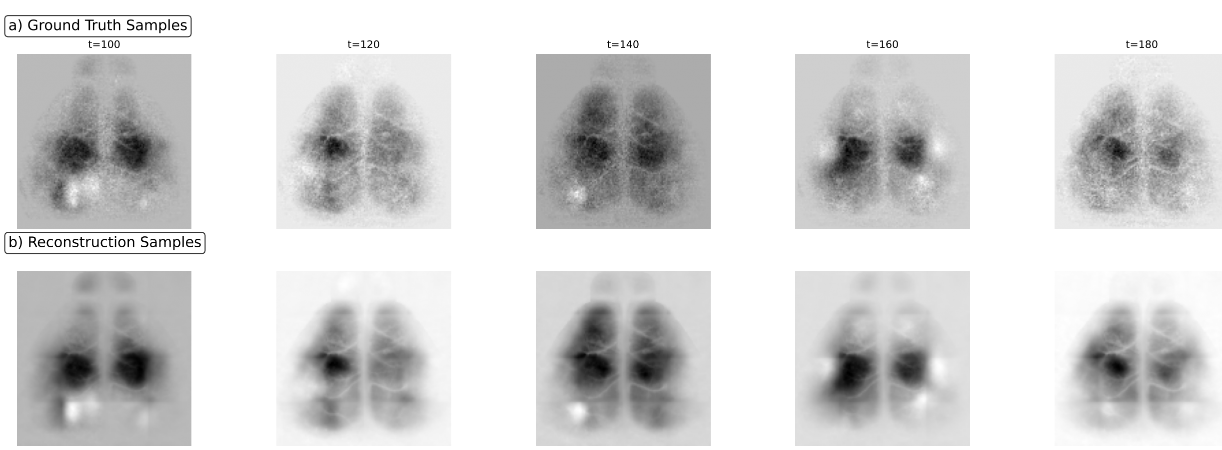

Figure 15. **Qualitative snapshots of neural reconstruction with masked regions.** Snapshots of **(a)** widefield activity (top row) across multiple time frames compared to **(b)** model predictions (bottom row). In these examples, all visual area (VIS) patches are masked (as indicated by the blue bounding boxes in Fig. 14, column VIS). Results are shown for a representative zero-shot (held-out) subject. Despite this, WiCAT accurately reconstructs the missing activity, demonstrating its ability to perform cross-region neural prediction from the available context.

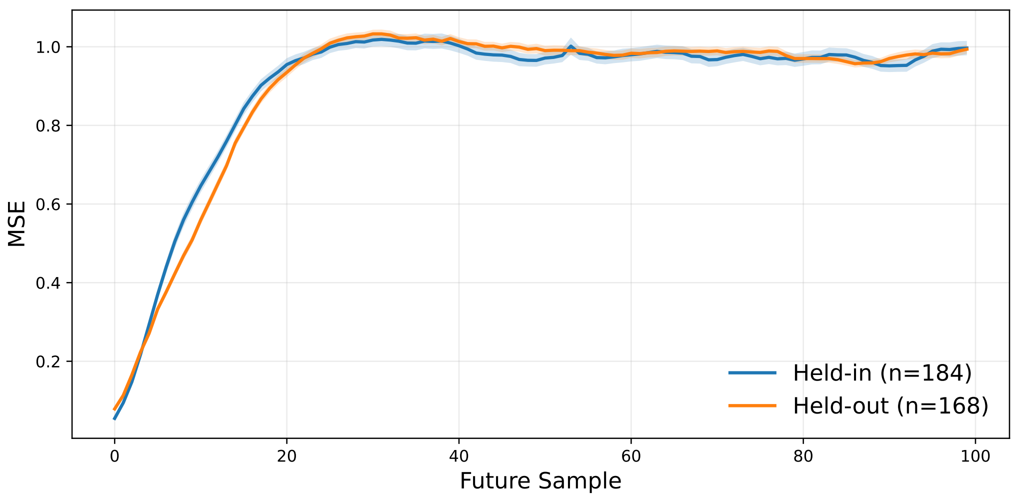

Figure 16. **Temporal prediction without future context.** Mean squared error (MSE) for predicting future frames when only the first part of the trial is provided as input (last 100 frames predicted without context), evaluated across held-in (pretraining set) and held-out (finetune and zero-shot sets) subjects. The results show that WiCAT preserves temporal structure and remains robust when generalizing to unseen subjects.

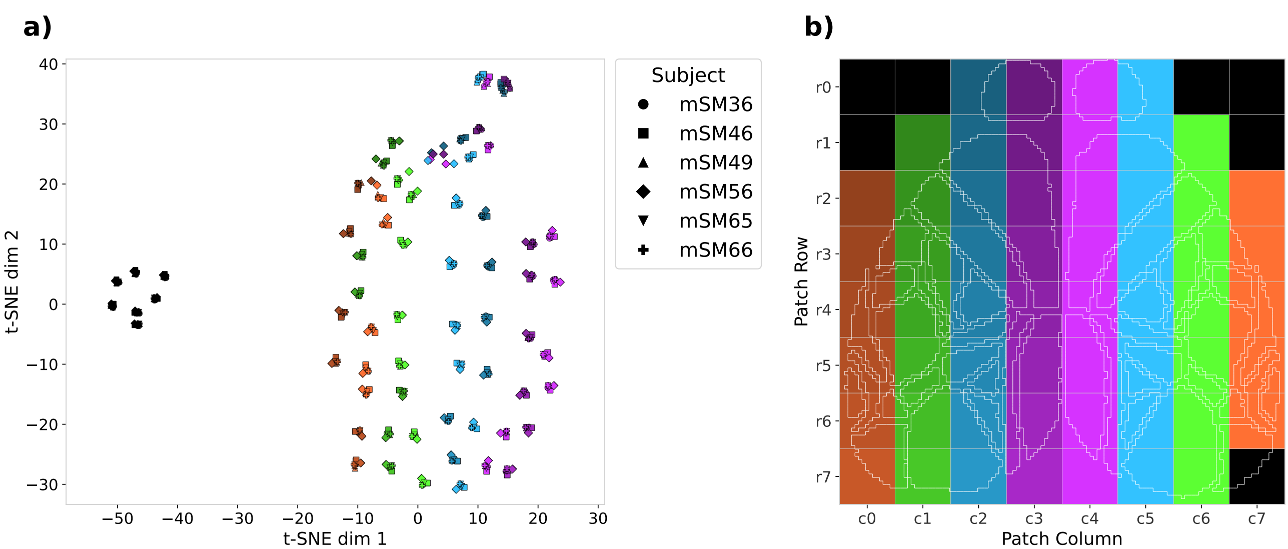

Figure 17. **t-SNE visualization of learned patch embeddings.** **(a)** Each point represents the mean embedding of a spatial patch (averaged over time and trials) for a given subject. Colors correspond to patch location, and markers denote subjects. Patches from the same spatial location across different subjects cluster together, while neighboring patches are embedded nearby, revealing preserved anatomical topology and indicating that the learned representations are subject-invariant. A clear left–right hemispheric symmetry is also observed, suggesting that corresponding regions across hemispheres are encoded similarly, potentially reflecting functional connectivity between bilateral brain regions. **(b)** Coloring scheme for the patch embeddings scattered in (a).

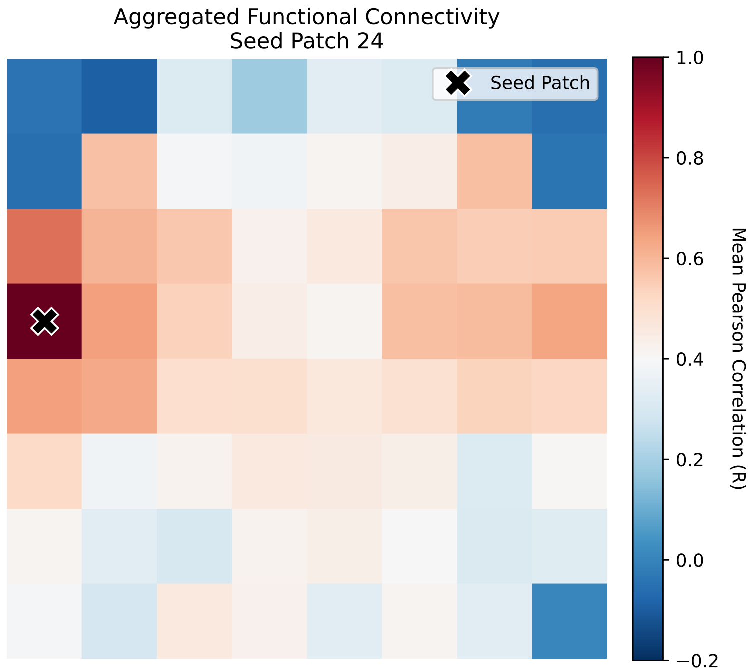

Figure 18. **Correlation structure of learned patch embeddings.** Mean pairwise correlations between patch embeddings, aggregated across sessions. The seed patch (marked) shows the highest correlation with neighboring regions and homologous areas across hemispheres. This structured correlation pattern further shows that the learned representations capture spatially coherent and anatomically meaningful relationships.

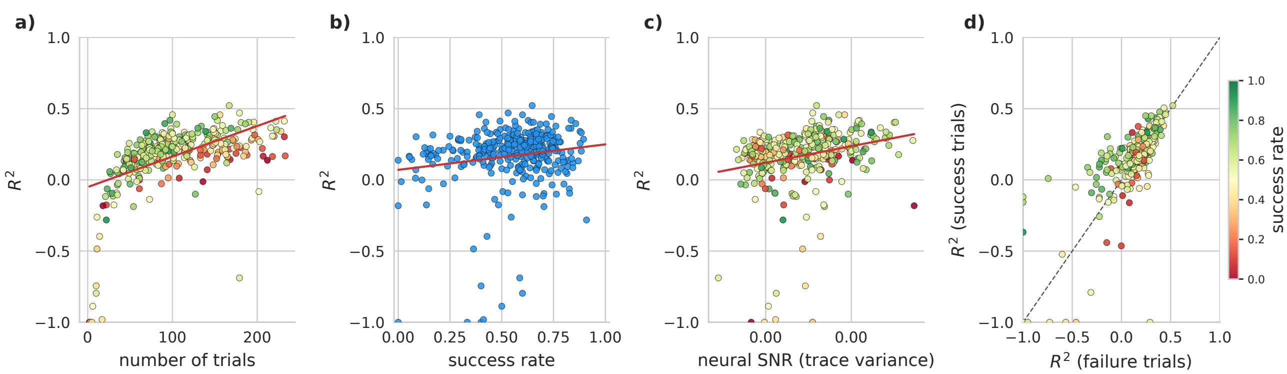

Figure 19. **Session-level factors affecting decoding performance (Kondo dataset).** Decoding performance **(a)** (R^2) vs. number of trials per session (color indicates success rate). **(b)** R^2 vs. session success rate, reflecting behavioral engagement. **(c)** R^2 vs. neural SNR (mean PCs trace variance from PCA). **(d)** R^2 for successful vs. failed trials (color indicates success rate; dashed line is identity). Across sessions ($n = 352$), trial count ($r = 0.52$, $p < 0.001$), neural variance as a proxy for neural SNR ($r = 0.246$, $p < 0.001$), and success rate as a proxy for task engagement ($r = 0.17$, $p = 0.002$) are positively correlated with decoding performance. Sessions with higher engagement cluster above the identity line in (d), indicating better decoding during successful trials.